codesprint

NATIONAL CODING COMPETITION

Bootcamp Task

Post-Secondary Category 2026

POWERED BY

mdia MALTA DIGITAL INNOVATION AUTHORITY

GOVERNMENT OF MALTA
MINISTRY FOR EDUCATION AND SPORTS
DIRECTORATE FOR STEM AND VET PROGRAMMES

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Hello!

The aim of this bootcamp task is to make sure that you're prepared for the post-secondary final challenge! The sort of tasks you'll be doing to complete this task will help you practice programming approaches that you'll be using for the final task. This gives you a chance to hone your skills!

Good luck!

Bootcamp Schedule

- Task Analysis (10 min)
- Task Development (60 min)
- Discussion and Demos (50min)

Loan Desk

The campus equipment loan desk tracks laptops, cameras, and robotic kits for students. Staff members need a way to monitor items on the shelf versus items out on loan. Each type of equipment has several details, some common to all and some specific to the device.

You are creating the software for this desk. The application will function as a catalogue that allows staff members to view, add, and lend out equipment. Your task is to build a Java or Python program to manage this catalogue. You may use a text-based interface or a graphical one. The application should load information from a file when it starts and saves data when requested.

Stay sharp, think carefully, and make the most of this bootcamp session.

Functionality #1: Program Initialisation

On start-up the system should:

– load the catalogue from a data file.

⚠ **If the file is missing or unreadable, start with an empty catalogue and notify the user.**

– Once loaded, the records must remain accessible to the system so they can be viewed, modified, or updated throughout the program's execution without needing to reload the file for every individual action (*see Functionality 2*).

– display a main menu with these options:

1. View catalogue
2. Add Item
3. Borrow / Return Item
4. Save Catalogue
5. Exit

⚠ **The program will only stop if the user chooses the 'Exit' option from the main menu.**

💡 **A text-based sample visual is shown in Screenshot 1.**

```
Loading catalogue from 'loanDesk_catalogue.csv' ...
10 records loaded successfully.

      =============== LOAN DESK SYSTEM ===============

                         Main Menu

 1. View catalogue
 2. Add Item
 3. Borrow / Return Item
 4. Save Catalogue
 5. Exit


Enter your choice (1-5):
```

Screenshot 1: Visual sample of program initialisation

Functionality #2: Records Structure

- Every item record shares four common fields:
  - unique ID,
  - name,
  - type, and
  - whether it is currently available.
- Each type includes its own specific fields:
  - Laptop: 'Operating System' and 'RAM (GB)'
  - Camera: 'Megapixels' and 'List of Lenses'
  - Robotics Kit: 'Piece Count' and 'Recommended Age Range'

💡 **A sample catalogue list is provided in Table 1 and as a CSV file on** https://bit.ly/cs26psec_loandesk.

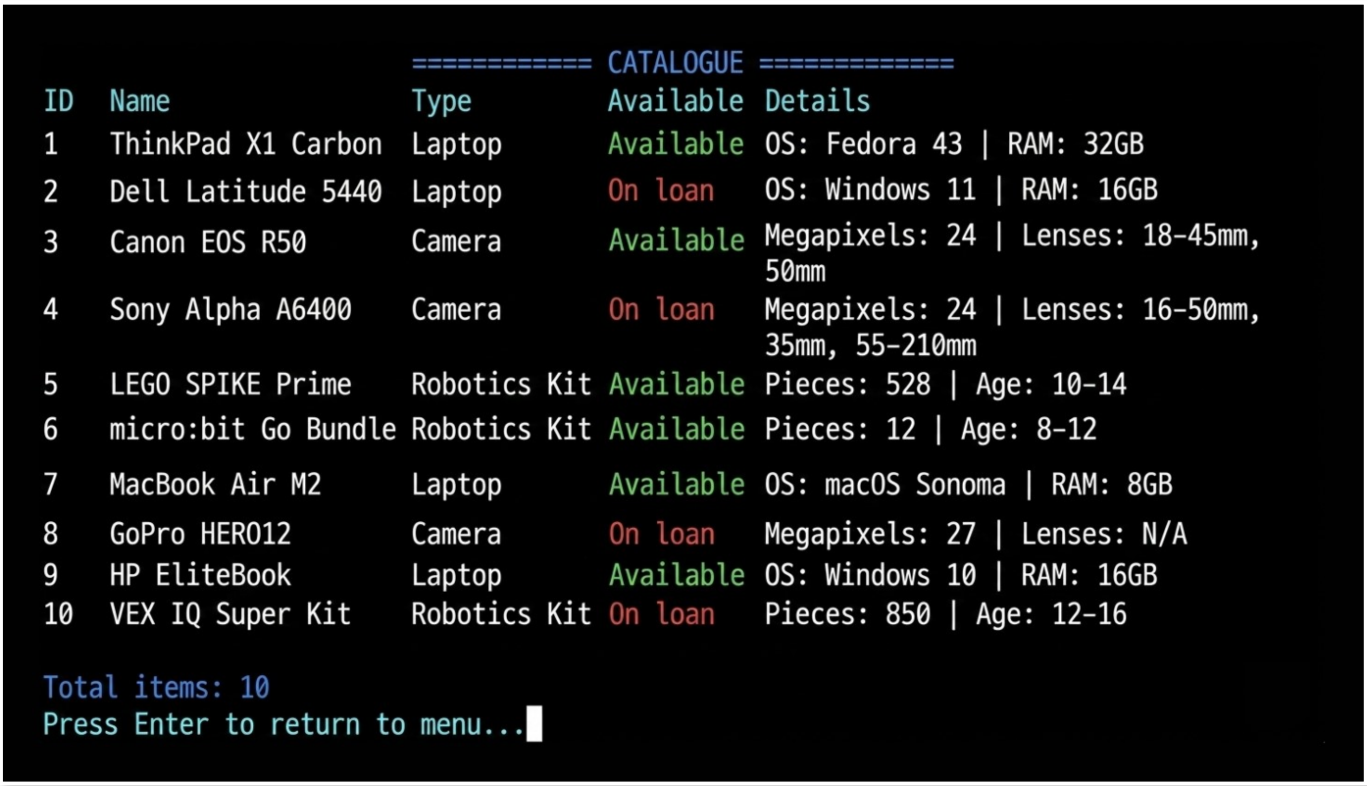
Functionality #3: View Catalogue

- List every item in a clear, aligned layout.
- Show the common fields and the type-specific details (for a camera, list its lenses).
- Show availability clearly (Available or On loan)

ID	Name	Type	Available	Specific Field 1	Specific Field 2
1	ThinkPad X1 Carbon	Laptop	Yes	OS: Fedora 43	RAM: 32GB
2	Dell Latitude 5440	Laptop	No	OS: Windows 11	RAM: 16GB
3	Canon EOS R50	Camera	Yes	Megapixels: 24	Lenses: 18-45mm, 50mm
4	Sony Alpha A6400	Camera	No	Megapixels: 24	Lenses: 16-50mm, 35mm, 55-210mm
5	LEGO SPIKE Prime	Robotics Kit	Yes	Pieces: 528	Age: 10-14
6	micro:bit Go Bundle	Robotics Kit	Yes	Pieces: 12	Age: 8-12
7	MacBook Air M2	Laptop	Yes	OS: macOS Sonoma	RAM: 8GB
8	GoPro HERO12	Camera	No	Megapixels: 27	Lenses: N/A
9	HP EliteBook	Laptop	Yes	OS: Windows 10	RAM: 16GB
10	VEX IQ Super Kit	Robotics Kit	No	Pieces: 850	Age: 12-16

Table 1: List of preloaded records

💡 **A text-based sample visual is shown in Screenshot 2.**



Screenshot 2: Visual sample of the 'Viewing Catalogue' option.

Functionality #4: Add an Item

- Prompt for which type of item to add.
- Prompt for the item's name, then the fields for that type (*see Functionality #2*).
- Assign a new unique ID automatically.
- New items are assigned as Available.
- Validate every entry (*see Functionality #7*):
  - names must not be empty;
  - RAM, megapixels and piece count must be whole numbers greater than zero;
  - lenses are entered as a comma-separated list, or N/A if the camera does not use interchangeable lenses.

Functionality #5: Borrow / Return

- Prompt for an item ID.
- If an item with that ID exists, flip its availability (Available ↔ On loan) and confirm the change.
- If an item with that ID is not found, display an appropriate message.

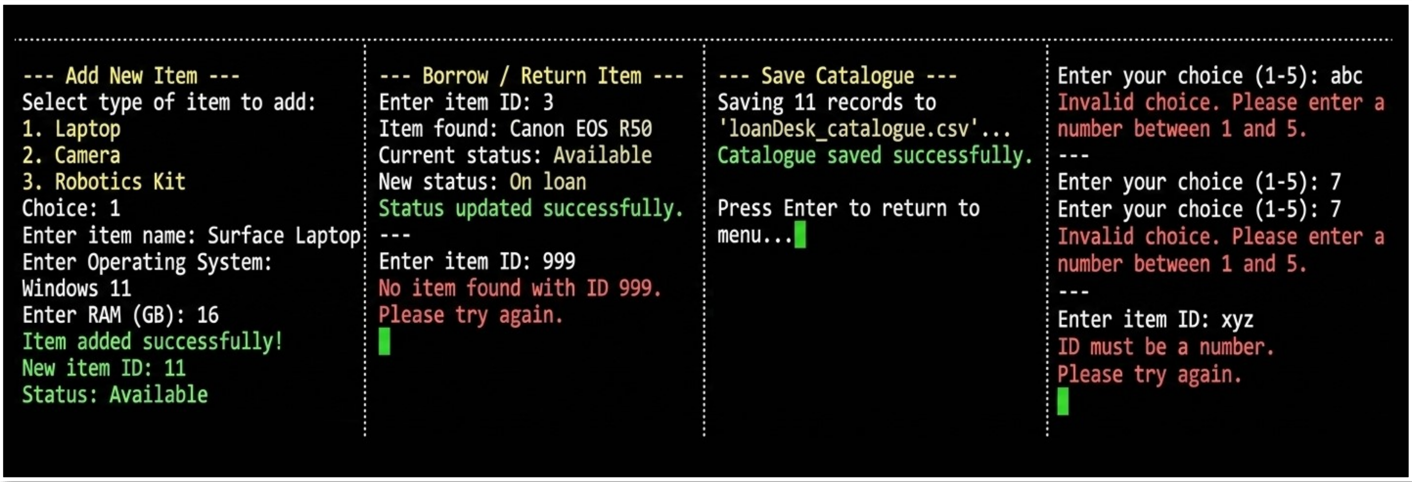
Functionality #6: Save Catalogue

- Write the whole catalogue back to the data file.
- Each record should include all the fields (common and specific ones) so the exact catalogue can be rebuilt next time it loads.

💡 **A round-trip rule applies, in that saving and then loading must reproduce the same catalogue.**

📝 **You may choose how to save the records; the file, or files, can be of any type, such as text file, JSON file, csv file etc.**

💡 **Some other text-based sample visuals are shown in Screenshot 3.**



Screenshot 3: Other visual samples.

Functionality #7: Check Routines

The system should:

- Avoid runtime errors.
- Treat user input (menu choices, item type input) as case insensitive.
- Invalid menu choices, non-numeric or unknown item IDs, and invalid field values are rejected, and the user is asked again.
- Numeric fields (RAM, megapixels, piece count) cannot be zero or negative.
- The program must treat a missing or corrupt data file as an empty catalogue and continue running without crashing.

Functionality #8: Modular Programming

- Structure your code using user-defined functions.
- Use dictionaries, lists, tuples, arrays and/or ArrayLists where appropriate.
- Include comments and apply Java or Python naming conventions.

FOR PYTHON:

Name your program **main_app.py,**

include all files into a folder called **postSec_loanDesk,**

*and submit as a Zip file named **postSec_name_surname**.*

FOR JAVA:

Name the class containing the main method **RunApp,**

include all files into a folder called **postSec_loanDesk,**

*and submit as a Zip file named **postSec_name_surname**.*

Assessment Rubric

Core Functionality (16 points)	
Program Initialisation & Menu	Shows the main menu; the program ends only via the Exit option.
Loading from File	Reads the data file and rebuilds each saved record as the correct item type.
Item Data Model	Each type carries its own extra fields, using classes or nested data.
View Catalogue	Clear listing showing common fields, type-specific details and availability.
Add Item	Adds an item of any type with a new unique ID; new items start available.
Borrow / Return	Toggles an item's availability, located by its ID.
Saving to File	Writes every item with its type and fields so the catalogue rebuilds exactly.
Nested Data Handling	The camera's list of lenses is stored, displayed and saved correctly.
Technical Implementation (8 Points)	
Input Validation	Menu, IDs and field entries are checked; input is not case sensitive.
Value Constraints	Numeric fields cannot be zero or negative; names cannot be empty.
Avoidance of Runtime Errors	Does not crash on bad input or a missing / corrupt data file.
Modular Programming & Structure	Well-organised classes and/or functions and suitable data structures.
User Experience (6 Points)	
Clear Listing & Formatting	The catalogue is easy to read and neatly aligned.
Python/Java Conventions & Comments	Appropriate naming conventions (camelCase / snake_case), meaningful names, consistent indentation and helpful comments.
File Naming	The program file is named according to the required format.
0 – Not Satisfactory | 1 – Partially Satisfactory | 2 – Entirely Satisfactory	
Maximum Score: 30 + 2 for every extra feature *	

* Extra features should extend the task functionality or demonstrate technical creativity WITHOUT altering or interfere with the core rules or required functionality of the task.

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